

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks Available |     |     |       |                    |
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| 7        |  |  |  | <p><b><u>Indicative content</u></b></p> <p><b>Eutrophication</b></p> <ul style="list-style-type: none"><li>• The nitrogen in the food fed to the fish may be lost as uneaten food, in faeces and as ammonia.</li><li>• The extra N and P in the water can be used by algae to grow. This algal bloom at the water surface can block light to the aquatic plants in the deeper water.</li><li>• With no light, these plants cannot photosynthesise and so die.</li><li>• Decomposers (bacteria and fungi) will then decompose the dead organic matter. They use (aerobic) respiration and so use up oxygen from the water.</li><li>• The water becomes deoxygenated and fish and other oxygen requiring species die.</li><li>• Anaerobic bacteria may start to reduce nitrates (denitrification)</li></ul> <p><b>Other impacts of fish farming</b></p> <ul style="list-style-type: none"><li>• The farmed fish tend to be packed tightly into a small area.</li><li>• This can lead to diseases passing through the population quickly and these can spread to the local wild fish population.</li><li>• To keep the stock healthy antibiotics are used. This can lead to antibiotic resistant bacteria developing.</li><li>• Pesticides used to kill the parasites may also be toxic to some of the local marine invertebrates.</li><li>• Farmed fish may have a selective advantage over wild species</li></ul> <p><b>Methods of preventing overfishing</b></p> <ul style="list-style-type: none"><li>• Net mesh sizes may be restricted. Larger mesh sizes allow immature fish to escape and they can go on to interbreed.</li><li>• Closed seasons for fishing are also enforced. These will be at times of the year when fish are breeding so again the stocks are replenished.</li><li>• Quotas - agreements which limit the catches brought ashore.</li><li>• Around the coast there are exclusion zones where fishing is not allowed. These areas allow fish to reproduce without being caught.</li><li>• Landing size regulations have been introduced which only allowing fish of a certain size to be caught. This allows some fish to return for breeding. Fishing of non-traditional varieties e.g. coalfish, has allowed stocks of other fish to recover.</li></ul> | 5               | 4   |     | 9     |                    |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks Available |     |     |       |         |         |
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|          |  |  |  | <p><b>7-9 marks</b><br/>Detailed explanation of eutrophication<br/><b>and</b><br/>Detailed explanation of other impacts of fish farming<br/><b>and</b><br/>Detailed explanation of methods of preventing overfishing<br/><i>The candidate constructs an articulate, integrated account, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i></p> <p><b>4-6 marks</b><br/>Any two from<br/>Explanation of eutrophication<br/>Explanation of other impacts of fish farming<br/>Explanation of methods of preventing overfishing<br/><i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i></p> <p><b>1-3 marks</b><br/>Brief explanation of eutrophication<br/><b>or</b><br/>Brief explanation of other impacts of fish farming<br/><b>or</b><br/>Brief explanation of methods of preventing overfishing<br/><i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i></p> <p><b>0 marks</b><br/><i>The candidate does not make any attempt or give a relevant answer worthy of credit</i></p> |                 |     |     |       |         |         |
|          |  |  |  | Question 7 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5               | 4   | 0   | 9     | 0       | 0       |